AquaSentry Universal All-In-One Operations Guide

Complete Step-by-Step Instructions: From Unzipping to Full Execution

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1. Welcome to AquaSentry

You have received the consolidated **AquaSentry Universal All-In-One Distribution**. This package contains everything required to analyze electrochemical water quality traces, run real-time artificial neural network classifications, host a full-stack water safety monitoring dashboard, deploy edge firmware to microcontrollers, and perform AI toxicology research.

What is Inside This Folder

When you open this folder, you will see the following layout: * **`START_HERE_OPERATIONS_GUIDE.pdf`**: *This document.* * **`setup_and_run.sh`**: 1-Click launcher for macOS & Linux. * **`setup_and_run.bat`**: 1-Click launcher for Windows. * **`run_all_tests.sh`** / **`run_all_tests.bat`**: 1-Click automated test runners. * **`main.py`**: The universal master command-line interface. * **`requirements.txt`**: Consolidated Python dependencies. * **`AquaSentry_Platform/`**: Full-stack FastAPI web application, interactive HTML5 dashboard, database, and serial bridge. * **`Neural_Network/`**: Dedicated AquaNet-V1 ML engine (Python, JavaScript, C99 microcontroller source, parity test suite). * **`AI_Agent/`**: LLM research agent for toxicology and environmental safety queries. * **`Data/`**: Real and synthetic square-wave ASV voltammetric scan CSV files and calibration matrices. * **`Analysis/`**: Signal processing, baseline subtraction, and data generation utilities. * **`Docs/`**: Full engineering blueprints, catamaran hardware specs, and technical reports.

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2. 3-Minute Quick Start on Any New Computer

System Requirements

- **Operating System**: macOS, Windows 10/11, or Linux (Ubuntu, Debian, Fedora, Arch).
- **Python**: Python 3.9, 3.10, 3.11, or 3.12 installed on your system. *(To verify, open terminal or command prompt and run `python3 --version` or `python --version`).*

Option A: macOS or Linux (Recommended 1-Click)

1. Open your **Terminal** application.

2. Navigate to this unzipped folder: `bash cd path/to/AquaSentry_Universal`
3. Run the automated launcher: `bash chmod +x setup_and_run.sh ./setup_and_run.sh`
4. **What happens automatically:**
5. The script detects your Python version.
6. It creates a clean, isolated virtual environment (`.venv`) customized for your exact operating system and CPU architecture.
7. It installs all dependencies from `requirements.txt` .
8. It starts the AquaSentry web server at `http://localhost:8000` and automatically opens your web browser.

Option B: Windows (Recommended 1-Click)

1. Open the unzipped folder in **Windows File Explorer**.
2. **Double-click** the file named: `bat setup_and_run.bat` *(Or open Command Prompt / PowerShell, navigate to the folder, and type `setup_and_run.bat`).*
3. **What happens automatically:**
4. Windows detects Python, creates `.venv` , installs dependencies, launches the server, and opens `http://localhost:8000` in your default browser.

Option C: Manual Setup (For Advanced Developers)

If you prefer complete manual control over your terminal environment:

```
# 1. Create a virtual environment
python3 -m venv .venv

# 2. Activate it
# On macOS / Linux:
source .venv/bin/activate
# On Windows (cmd):
.venv\Scripts\activate.bat
# On Windows (PowerShell):
.venv\Scripts\Activate.ps1

# 3. Install packages
pip install --upgrade pip
pip install -r requirements.txt

# 4. Launch the web platform
python main.py --serve --port 8000
```

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3. How to Use the Web Platform & Interactive Dashboard

Once the server is running (`http://localhost:8000`):

1. **Dashboard Overview**:
2. View live hydrogeochemical parameters: Arsenic ($[As]$), pH, Electrical Conductivity (EC), Turbidity, and Temperature.
3. View the real-time safety potability classification (`Safe`, `Caution`, `Unsafe`, `Hazardous`).
4. **Upload & Process Voltammetric Trace (ASV)**:
5. In the UI, navigate to the **Voltammetry** tab.
6. Upload any scan file from the `Data/` folder (such as `Data/example_data.csv`).
7. The system applies a 5-point Savitzky-Golay filter, removes baseline tilt, identifies the net Faradaic stripping peak, inverts calibration to find $[As]$, and passes it through the AquaNet neural network.
8. **Interactive AI Safety Assistant**:
9. Open the **Chat** widget in the bottom right corner.
10. Ask questions such as: *"Is this water safe to drink?"*, *"What does 0.035 mg/L arsenic mean?"*, or *"Explain WHO guidelines."*
11. The assistant enforces emergency toxicology guardrails and warns against drinking contaminated water.
12. **Interactive REST API Documentation (Swagger)**:
13. Visit `http://localhost:8000/docs` to view and execute any of the 15+ interactive REST endpoints.

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4. Running the Complete Verification Test Suite

To verify that every component, mathematical calculation, and safety rule works with 100% precision:

On macOS / Linux:

```
./run_all_tests.sh
```

On Windows:

```
run_all_tests.bat
```

Or using the Universal CLI:

```
python3 main.py --test
```

What is Tested (97 Automated Test Cases):

- **18 Standalone Neural Parity Tests** (`Neural_Network/test_aquanet_parity.py`):
- Exact 23-node, 112-synapse forward pass parity.
- Multi-parameter potability downgrades (pH outside 6.5–8.5, turbidity $> 5.0\text{ NTU}$, conductivity $> 1500\ \mu\text{S/cm}$).
- Out-of-bounds sensor fail-safes (invalid negative temperatures or extreme pH immediately revoke safe status).
- Conservative statutory precedence overrides (arsenic $>0.150\text{ mg/L} \to \text{Hazardous}$; $\ge 0.050\text{ mg/L} \to \text{Unsafe}$).
- **79 Platform Integration Tests** (`AquaSentry_Platform/run_tests.py`):
- REST API contracts, reading ingestion, duplicate protection.
- Digital filtering, Savitzky-Golay convolution, peak detection.
- Chat policy prompt injection prevention.

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5. Universal Command-Line Interface (`main.py`)

You can execute common tasks directly from your terminal using `main.py`:

Command	Action
`python3 main.py --serve`	Starts the web platform and API on port 8000.
`python3 main.py --serve --port 8080`	Starts on custom port 8080.
`python3 main.py --test`	Runs all 97 test cases across both suites.
`python3 main.py --csv Data/example_data.csv`	Processes a raw voltammetric CSV file and prints classification JSON.
`python3 main.py --csv Data/example_data.csv --ph 7.8 --ec 500 --turb 3.0 --temp 24`	Processes CSV with specific field sensor values.
`python3 main.py --neural`	Runs the standalone AquaNet neural network benchmark scenarios.
`python3 main.py --version`	Displays architecture topology and installed versions.

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6. Embedded Microcontroller Deployment (C99 Engine)

If you are deploying AquaSentry on embedded hardware (ESP32, STM32, Arduino, or ARM Cortex-M):

1. The C99 files are located in:
2. Header: `Neural_Network/aquanet_v1.h`
3. Implementation: `Neural_Network/aquanet_v1.c`
4. **Compilation**: The engine is written in standard ANSI C99 with **zero dynamic memory allocations** (`malloc`). It compiles with zero warnings: `bash cd Neural_Network clang -std=c99 -Wall -Wextra -Werror -c aquanet_v1.c`
5. **Firmware Integration Example**: ```c #include "aquanet_v1.h"

AquaNetReading reading = { .arsenic_mg_l = 0.0074f, .ph = 7.2f, .conductivity_us_cm = 400.0f, .turbidity_ntu = 2.0f, .temperature_c = 24.0f };

AquaNetResult result; aquanet_predict(&reading, &result);

// result.predicted_class: 0 = Safe, 1 = Caution, 2 = Unsafe, 3 = Hazardous // result.confidence_pct: Confidence percentage (0.0 to 100.0) ```

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7. Standards Compliance & Potability Limits

The platform adheres to international drinking water health standards:

Standard / Parameter	Threshold	Safety Enforcement
WHO Arsenic Guideline	0.010 mg/L ($10\ \mu\text{g/L}$)	Caution triggered if exceeded.
BIS 10500 Permissible	0.050 mg/L ($50\ \mu\text{g/L}$)	Hard override forces `Unsafe`. Downgrades prohibited.
Acute Toxic Hazard	$> 0.150\text{ mg/L}$ ($150\ \mu\text{g/L}$)	Hard override forces `Hazardous`.
Method Limit of Detection	0.020 mg/L ($20\ \mu\text{g/L}$)	Analytical notice attached to output.
Drinking Water pH	$6.5 - 8.5$	Violations trigger potability downgrade.
Drinking Water Turbidity	$\le 5.0\text{ NTU}$	$> 5.0\text{ NTU}$ triggers potability downgrade.
Conductivity (TDS)	$\le 1500\ \mu\text{S/cm}$	$> 1500\ \mu\text{S/cm}$ triggers potability downgrade.

8. Troubleshooting & FAQs

- **Q: The terminal says `python3: command not found` or `python: command not found`.**
  Fix: Install Python 3.9+ from python.org and ensure you check *"Add Python to PATH"* during installation on Windows.
- **Q: The server says `Port 8000 is already in use`.**
  Fix: Start on an alternative port by running: `python3 main.py --serve --port 8080`.
- **Q: On macOS/Linux, I get `permission denied: ./setup_and_run.sh`.**
  Fix: Make the script executable by running: `chmod +x setup_and_run.sh run_all_tests.sh`.
- **Q: How do I stop the web server?**
  Fix: Press `Ctrl + C` in the terminal window where the server is running.